

AutoTradeBot

How the app works - an architecture guide for a computer scientist

Repository: [omar3903/ClaudeAutoBot](#) (master, 16 September 2026)

Python 3.10 · FastAPI · SQLAlchemy · `ib_async` · pandas/numpy · plain ES-module dashboard

Contents: the twelve books · the big picture · the runtime · a play's life · the strategies · the data and the databases · the research loop and the learning storage · the dashboard and the API · the important functions · a glossary of the trading terms

Companion: [AutoTradeBot-learning.pdf](#), the machine-learning plan in detail

1. The twenty books the app was trained on

The PDFs live in *OneDrive/Desktop/Training books & Documentation papers/Financial-Economic books/Used books/*. The first four gave the app its trade setups and the way it explains a trade; the next eight gave it the statistics behind the replay, the noise checks, the volatility model, the market regime and the pairs desk. The last eight (added 2026-09-18) are about not fooling yourself: whether a record is luck, what trading really costs, a model that learns which plays pay, and two more books of setups. The last column says where each idea lives in the code.

Title	Author	Edition	What it contributed	Where in the code
How to Day Trade for a Living	Andrew Aziz	self-published, 2016	the day-trade setups (opening-range breakout, VWAP, flags, ABCD, reversals), 2:1 reward-to-risk, the 2% rule, time-of-day weights, the daily maximum loss, gappers watchlist, scaling out at the first target	strategies/technical.py, analysis/levels.py, risk/position_sizing.py, execution/autopilot.py (daily loss), scanner/heat.py (gappers), execution/exit_manager.py (scale-out)
Trading in the Zone	Mark Douglas	Prentice Hall 2000 (Wiley 2021 printing)	thinking in probabilities: every play states an edge, its odds, and what invalidates it	strategies/base.py (the explanation, invalidation and odds fields on every Play)
Technical Analysis of the Financial Markets	John J. Murphy	New York Institute of Finance, 1999	horizontal support and resistance, oscillator divergence, trend reading	analysis/levels.py, indicators/ta.py (rsi_divergence, macd_divergence), the divergence_reversal and sr_bounce setups
Financial Modeling and Valuation: A Practical Guide to Investment Banking and Private Equity, 2nd ed.	Paul Pignataro	Wiley, 2022	enterprise value, comparable multiples, discounted cash flow, the football-field band	valuation/ (dcf.py, multiples.py, football_field.py, projections.py), strategies/fundamental.py, data/sec_edgar.py (the financials)
Quantitative Trading: How to Build Your Own Algorithmic Trading Business	Ernest P. Chan	Wiley, 2009	backtests must charge costs, keep a held-out sample, and paper-trade before trusting; half-Kelly sizing	research/replay.py (costs, held-out third), quant/sizing.py, research/weights.py (live trades count double)
Algorithmic Trading: Winning Strategies and Their Rationale	Ernest P. Chan	Wiley, 2013	stationarity tests (ADF, Hurst, variance ratio), half-life of mean reversion, buy-on-gap, post-earnings drift, momentum fails in turbulent markets	quant/stationarity.py, quant/readings.py, scanner/noise.py (not_trending, not_mean_reverting, turbulent_market), strategies/statistical.py
Analysis of Financial Time Series, 3rd ed.	Ruey S. Tsay	Wiley, 2010	GARCH(1,1) volatility forecasting, the market model	quant/volatility.py, quant/market_model.py (abnormal moves vs SPY)
Applied Econometric Time Series, 4th ed.	Walter Enders	Wiley, 2014	the same volatility toolkit (RiskMetrics EWMA) and unit-root testing	quant/volatility.py (ewma_var), quant/stationarity.py
Time Series Analysis	James D. Hamilton	Princeton University Press, 1994	chapter 22: Markov switching regimes fitted by EM - the calm/turbulent market pill	quant/regime.py, engine/market_regime.py (SPY, 3 years of daily returns)

Title	Author	Edition	What it contributed	Where in the code
Pairs Trading: Quantitative Methods and Analysis	Ganapathy Vidyamurthy	Wiley, 2004	spread construction, the nonparametric band that maximises profit per crossing, holding periods	quant/bands.py, pairs/model.py, pairs/desk.py
Likelihood-Based Inference in Cointegrated Vector Autoregressive Models	Søren Johansen	Oxford University Press, 1995	the Johansen trace test used to confirm a pair is cointegrated	quant/cointegration.py (johansen), pairs/finder.py
The Cointegrated VAR Model: Methodology and Applications	Katarina Juselius	Oxford University Press, 2007	how to read the cointegration rank in practice; Engle-Granger as the first pass	quant/cointegration.py (engle_granger), pairs/finder.py
Evidence-Based Technical Analysis	David R. Aronson	Wiley, 2006	a rule's record is a hypothesis test: bootstrap the mean against zero, test the best of many rules against the best that luck makes (White's reality check), and detrend first so being long in a rising market is not an edge	research/significance.py (luck_test, reality_check), research/replay.py (drift_r on every trade), execution/autopilot.py (proof_p_value in the proof rule)
Advances in Financial Machine Learning	Marcos López de Prado	Wiley, 2018	meta-labelling (the setups pick the side, a model decides whether to act and how big), sample weights by uniqueness with time decay, purged walk-forward folds with an embargo, out-of-sample feature importance, bet size from the predicted probability	research/model.py (Boosted, sample_weights, importance, bet_size), research/validate.py (the folds)
Machine Learning for Algorithmic Trading, 2nd ed.	Stefan Jansen	Packt, 2020	gradient boosting on tabular features with gaps, calibrated probabilities, the information coefficient of a feature, time-series cross-validation	research/model.py (HistGradientBoosting, isotonic calibration, information_coefficients), scripts/train_model.py
Trade Your Way to Financial Freedom, 2nd ed.	Van K. Tharp	McGraw-Hill, 2006	R-multiples and expectancy (the app's unit of account), the quality of an R-multiple distribution (the number he later named SQN), the marble-bag simulation of drawdowns, the percent-risk sizing model	research/significance.py (sqn, marble_bag), risk/position_sizing.py
Systematic Trading	Robert Carver	Harriman House, 2015	costs are the one number known in advance: a rule that pays more than a third of its pre-cost return in costs trades too fast (the speed limit)	research/replay.py (cost_r on every trade), research/significance.py (cost_share, SPEED_LIMIT), the proof rule

Title	Author	Edition	What it contributed	Where in the code
Trading and Exchanges: Market Microstructure for Practitioners	Larry Harris	Oxford University Press, 2003	implementation shortfall (the fill against the price at the decision), the spread as the price of immediacy, why a stale limit order is a free option for someone else	engine.py (_chase_check: spread gate, the quote kept at the decision), execution/executor.py (expire_entries), trades.entry_slippage_bps / exit_slippage_bps, research/journal.py (execution_quality)
The Art and Science of Technical Analysis	Adam H. Grimes	Wiley, 2012	market structure tested statistically; the two templates he trades: the failure test (Wyckoff's spring and upthrust) and the pullback after a momentum thrust, read with a 20-EMA inside 2.25-ATR Keltner channels	strategies/patterns.py (failure_test, trend_pullback)
Encyclopedia of Chart Patterns, 2nd ed.	Thomas N. Bulkowski	Wiley, 2005	measured statistics for chart patterns: twin bottoms fail 64% of the time until a close confirms them; confirmed, the projected height is met about two times in three and half throw back first	strategies/patterns.py (double_bottom: confirmation, measure rule, a stop inside the pattern)

Only Murphy and Hamilton are scanned images; the rest are text PDFs. None of the statistics needs SciPy or statsmodels: the models in `quant/` and `research/significance.py` are hand-written NumPy (OLS, Nelder-Mead, EM, the bootstrap). Only the learned model (`research/model.py`) uses scikit-learn, and the app runs without it. `scripts/r/audit_records.R` recomputes the record statistics in base R as an independent check; nothing live ever waits on R.

2. The big picture

What it is. AutoTradeBot is a single-process Python application that watches the US stock market through an Interactive Brokers (IBKR) connection, proposes trades ("plays") from a catalogue of strategies, optionally enters them on its own ("Autopilot"), always manages the exits, and keeps a record of everything in a database. A browser dashboard on `127.0.0.1:8787` shows the state and takes the operator's decisions.

In one sentence per layer:

- **Dashboard** (`tos_bot/web/`): static HTML plus ES modules; talks REST for actions and listens on a WebSocket for events.
- **Server** (`tos_bot/server/app.py`): a FastAPI app whose routes are thin wrappers over engine methods; `/ws` streams the event bus.
- **Engine** (`tos_bot/engine/`): the conductor. Owns every service, runs seven daemon threads, and is the only writer of the runtime state.
- **Scanner and strategies** (`scanner/`, `strategies/`): turn candles into plays on a schedule.
- **Execution** (`execution/`): Autopilot gates, the Executor sends orders, the ExitManager manages open positions.
- **Research** (`research/`, `quant/`): the replay (backtest), the daily journal, the evidence weights, and the statistical models.
- **Data and brokers** (`data/`, `brokers/`, `persistence/`): market data with an on-disk cache, the IBKR adapter and a simulator, and the SQL layer.

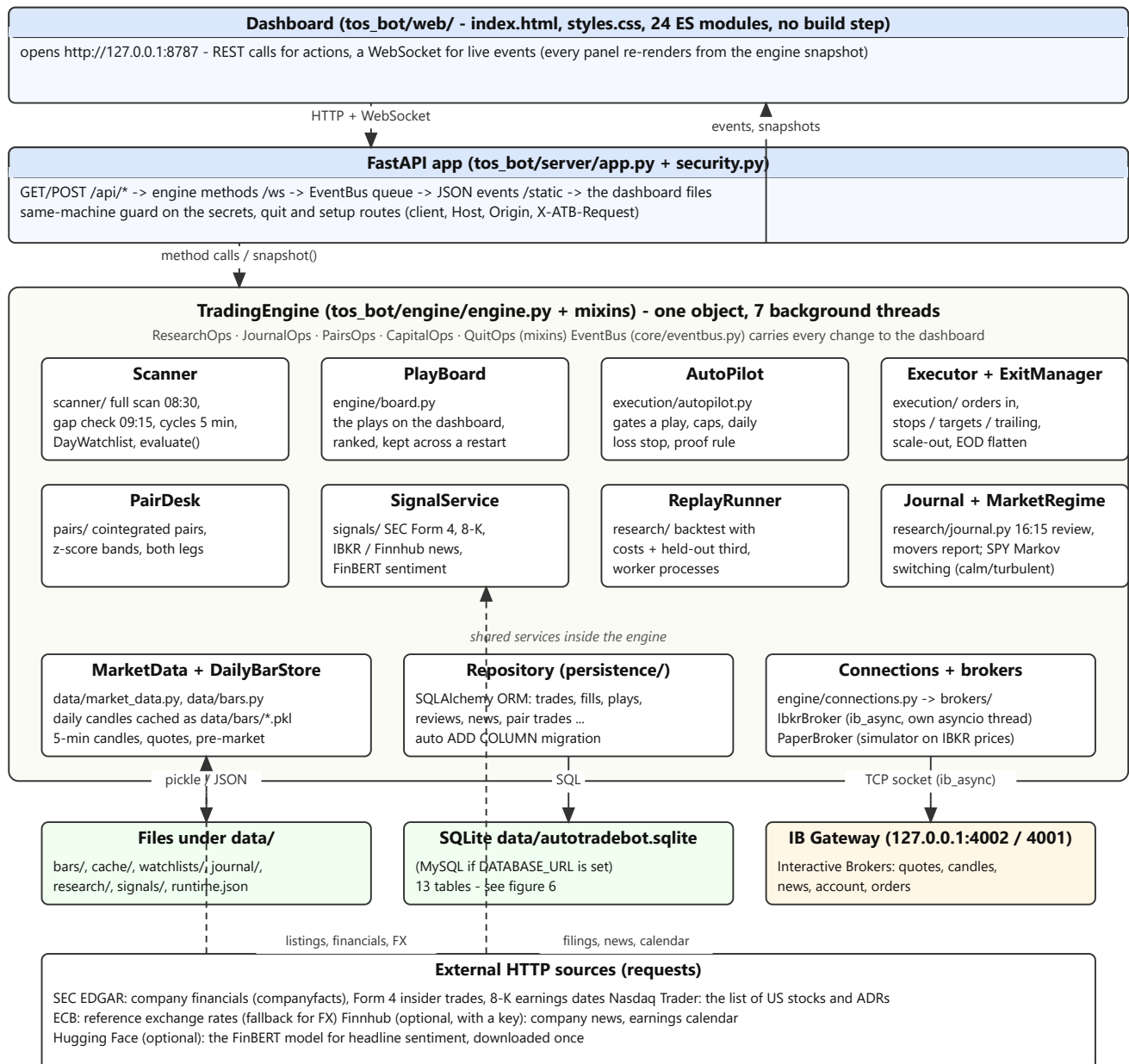


Figure 1 - Components: the browser, the FastAPI server, the engine and its services, and where data comes from and goes to.

The core classes

Figure 2 is the class view of the same thing. The engine is a composition root: it constructs every service in `__init__`, hands them what they need (the repository, the market data, the event bus) and exposes the operator actions the server calls. The mixin classes only split the engine's source into files by topic; at runtime there is one object. Strategies are stateless: a `StrategyContext` per stock carries the data and the memoised indicators, and `generate()` returns plays.

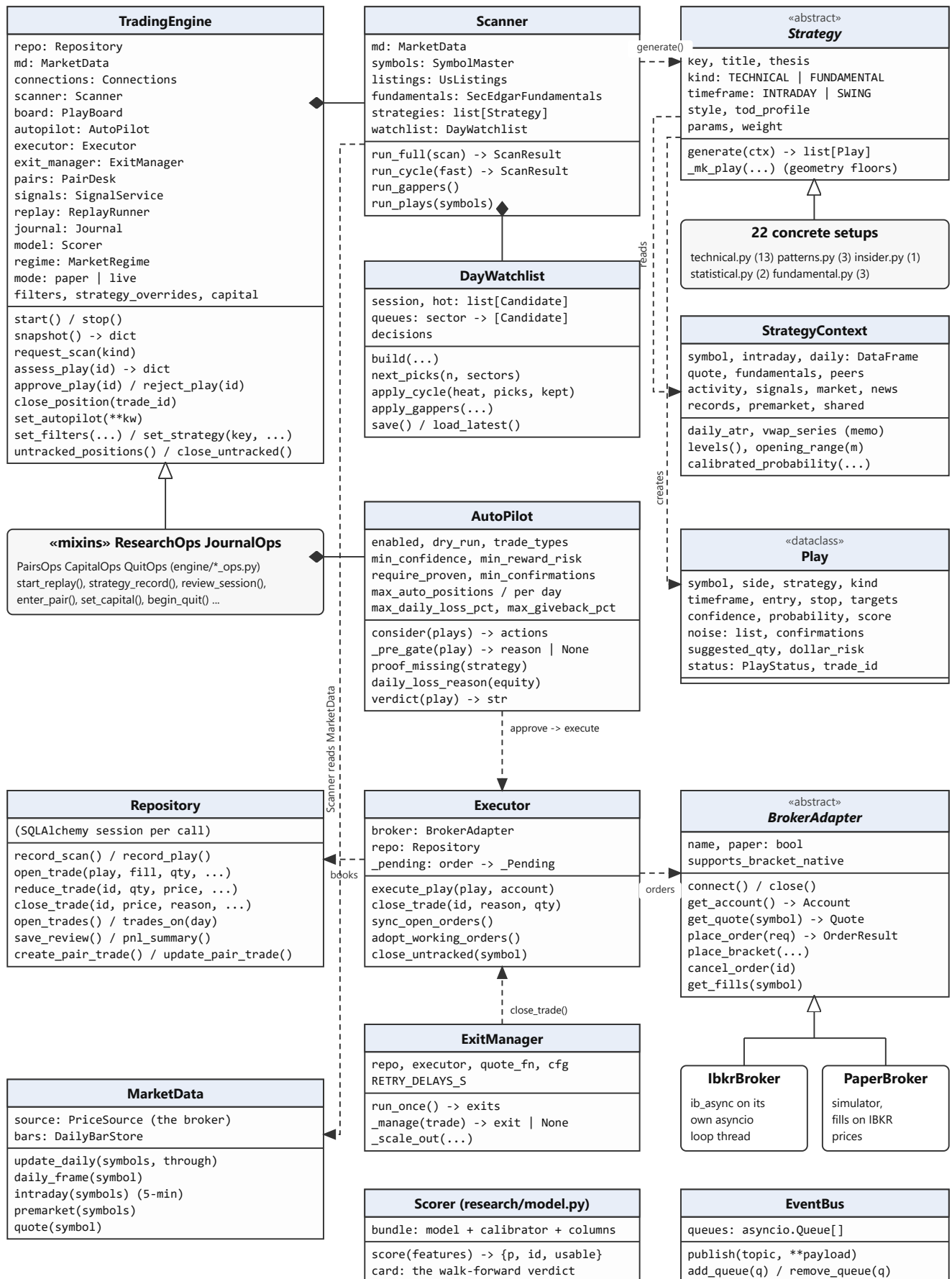


Figure 2 - Class diagram of the core: the engine and what it owns, the scanner and the strategies, the trade path, the broker interface and the persistence layer.

Technology stack

Concern	Choice	Notes
Language / runtime	Python 3.10, one process	Started by <code>run.py</code> ; <code>scripts/run_24_7.bat</code> keeps it up
Web server	FastAPI + uvicorn, WebSocket	REST routes call the engine in a thread pool; <code>security.py</code> restricts sensitive routes to the same machine
Broker / prices	ib_async against IB Gateway	Paper on port 4002, live on 4001. The only price source; delayed data is accepted
Simulator	<code>brokers/paper_adapter.py</code>	Fills on IBKR prices; its state is in <code>data/paper_state.json</code>
Database	SQLAlchemy 2 ORM; SQLite by default	MySQL/PyMySQL if <code>DATABASE_URL</code> is set; falls back to SQLite when MySQL is unreachable
Numerics	pandas, numpy	No TA-Lib, no SciPy: <code>indicators/ta.py</code> and <code>quant/</code> are vectorised by hand
Config	pydantic-settings	<code>.env</code> for secrets, <code>config/config.yaml</code> for defaults, <code>data/runtime.json</code> for dashboard choices (it wins)
Dashboard	Vanilla JS ES modules	No bundler, no framework; <code>state.js</code> is a small pub/sub store
Tests	pytest, 474 tests in 52 files	<code>tests/fakes.py</code> fakes the Gateway; <code>-m slow</code> boots the whole engine
Optional ML	transformers + torch (FinBERT)	Headline sentiment; the app runs without it
Optional learning	scikit-learn	The meta-label model (<code>research/model.py</code>): boosted trees and isotonic calibration; the app runs without it, with no model
Independent audit	base R (optional)	<code>scripts/r/audit_records.R</code> recomputes the record statistics offline; nothing live waits on R

How the repository is laid out

<code>run.py</code>	boot the engine + dashboard (Ctrl+C follows the quit rules)
<code>config/config.yaml</code>	defaults for every panel (<code>config.example.yaml</code> is the template)
<code>data/</code>	everything the app writes (git-ignored): SQLite, candle cache, journal ...
<code>tos_bot/</code>	
<code>config.py</code>	<code>.env</code> + <code>config.yaml</code> -> Settings (pydantic)
<code>core/</code>	enums, dataclasses (Play, Account, OrderRequest ...), EventBus
<code>engine/</code>	TradingEngine + mixins, board, connections, reconcile, runtime, regime
<code>scanner/</code>	schedule, heat, watchlist, evaluator, filters, noise, scanner
<code>strategies/</code>	base (StrategyContext, Strategy), technical, statistical, fundamental,
<code>insider</code>	
<code>execution/</code>	order_builder, executor, exit_manager, autopilot
<code>risk/</code>	position_sizing, pdt_guard
<code>research/</code>	replay, runner, history, weights, journal, movers
<code>quant/</code>	stationarity, cointegration, volatility, regime, bands, sizing,
<code>market_model</code>	
<code>pairs/</code>	model, finder, backtest, desk
<code>signals/</code>	service, store, book, form4, edgar, news, finnhub, sentiment, calendar
<code>data/</code>	market_data, bars, listings, symbols, sec_edgar, fundamentals, fx, sectors
<code>valuation/</code>	enterprise_value, multiples, dcf, projections, football_field
<code>analysis/</code>	levels (support/resistance), candles
<code>indicators/ta.py</code>	SMA/EMA/RSI/MACD/ATR/Bollinger/Keltner/ADX/VWAP/opening range/divergence
<code>brokers/</code>	base (BrokerAdapter), venues, ibkr_adapter, paper_adapter
<code>persistence/</code>	db (engine + migration), models_orm (tables), repository (queries)
<code>server/</code>	<code>app.py</code> (routes + WebSocket), <code>security.py</code>
<code>web/</code>	<code>index.html</code> , <code>styles.css</code> , <code>js/*.js</code>
<code>util/</code>	clock (NYSE calendar to 2028), logging, keep_awake, net
<code>tests/</code>	pytest (<code>fakes.py</code> = a synthetic Gateway)
<code>scripts/</code>	<code>init_db.py</code> (MySQL schema), <code>ibkr_setup.py</code> (Gateway doctor), <code>run_24_7.bat</code>

3. The runtime: process, threads and loops

`python run.py` builds the FastAPI app, which constructs one `TradingEngine` and calls `start()`. Start binds the broker chosen by the routing (`brokers/venues.py`: IBKR paper, IBKR live, or the simulator on IBKR prices), reads the account once, and launches the seven daemon threads below. Every thread sleeps on the same `threading.Event`, so `stop()` ends them together. Ctrl+C goes through the quit rules: in paper it closes positions and exits; in live with positions open it asks in the dashboard first.

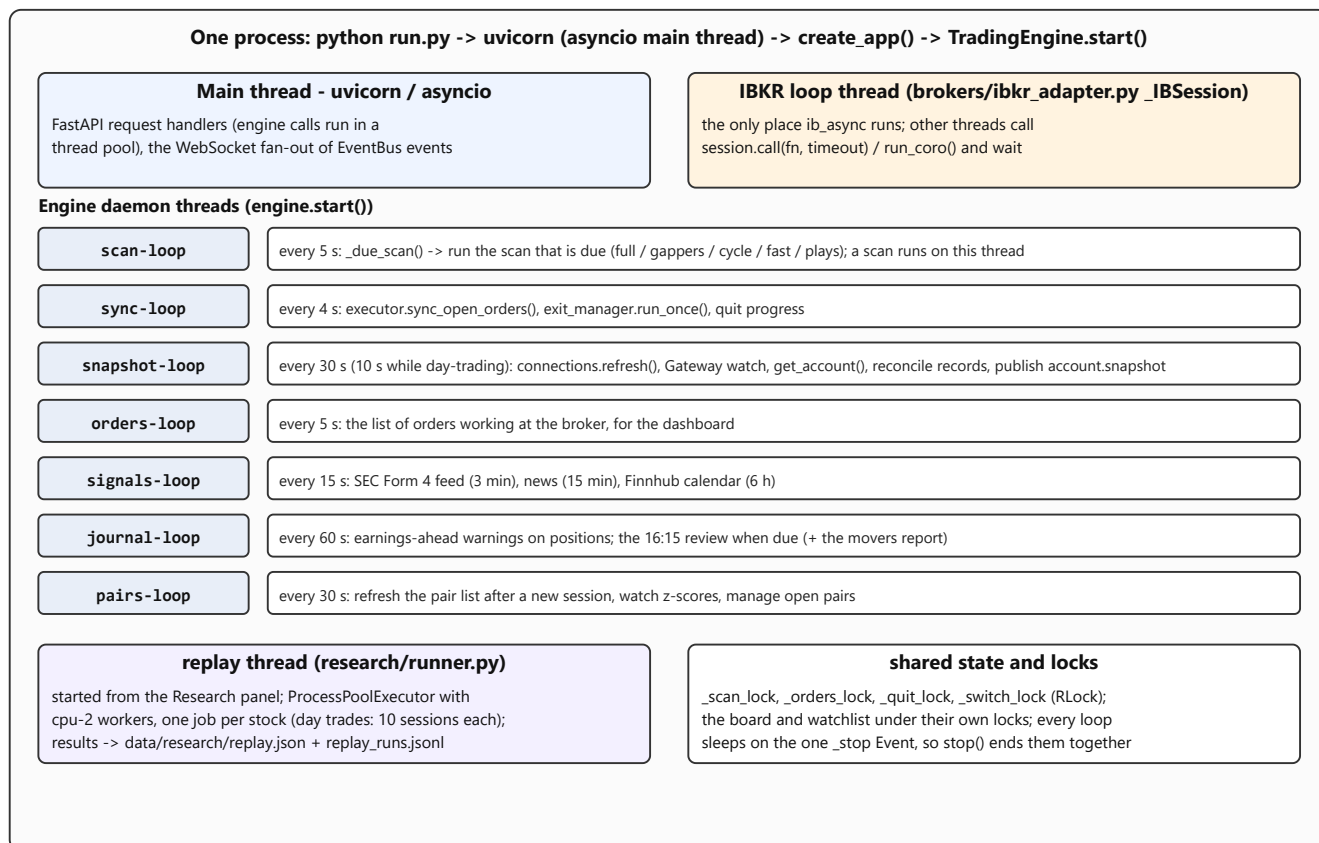


Figure 3 - Threads: everything runs in one Python process; the engine's loops are daemon threads, the replay uses worker processes.

The scan schedule (`scanner/schedule.py`)

Scan kind	When	What it reads	What it produces
full	once a day at 08:30 ET (settable 04:00-09:00); also on start-up if today's is missing	the Nasdaq listing of ~5,700 stocks and ADRs, IBKR contract details, one year of daily candles for every tradable name (top-ups after the first day), SEC financials for the leaders	a ranked universe by daily heat, the day's hot list (20) and per-sector buffers (25 each), swing plays for the top 400 names, valuation plays
gappers	09:15 ET (settable 08:00-09:25)	pre-market 5-minute candles of the hot list and buffers	swaps the coolest non-gapping hot names for stocks gapping 2%+ on 50k+ shares
cycle	every 5 minutes while the market is open	5-minute candles of the hot list + 2 buffer names per sector	day-trade and swing plays; adopt / keep / drop decisions on the buffers
fast	every 60 s while Autopilot is day-trading	the hot list only	fresh day-trade plays
plays	every 15 s	the stocks that already have plays on the board	re-evaluates them so stale plays leave

The design is deliberate: a full sweep of the market takes about ten minutes of Gateway time (each candle request takes the Gateway about half a second whatever its length; twelve in flight read ~17 stocks/s, and thirty-two time out). So the whole market is read once a day and the day's attention goes to a small, sector-diverse list. Swing setups read daily candles, which change once a day, so rescanning everything intraday would not find more of them.

4. The life of a play

A **play** is the app's unit of work: a proposed trade with a symbol, a side (LONG or SHORT), an entry price, a stop (where the idea is wrong), one or two targets, the setup's confidence and odds, noise flags, and a rank score. It is a plain dataclass (`core/models.py Play`). Figure 4 follows one from a scan cycle to its exit.

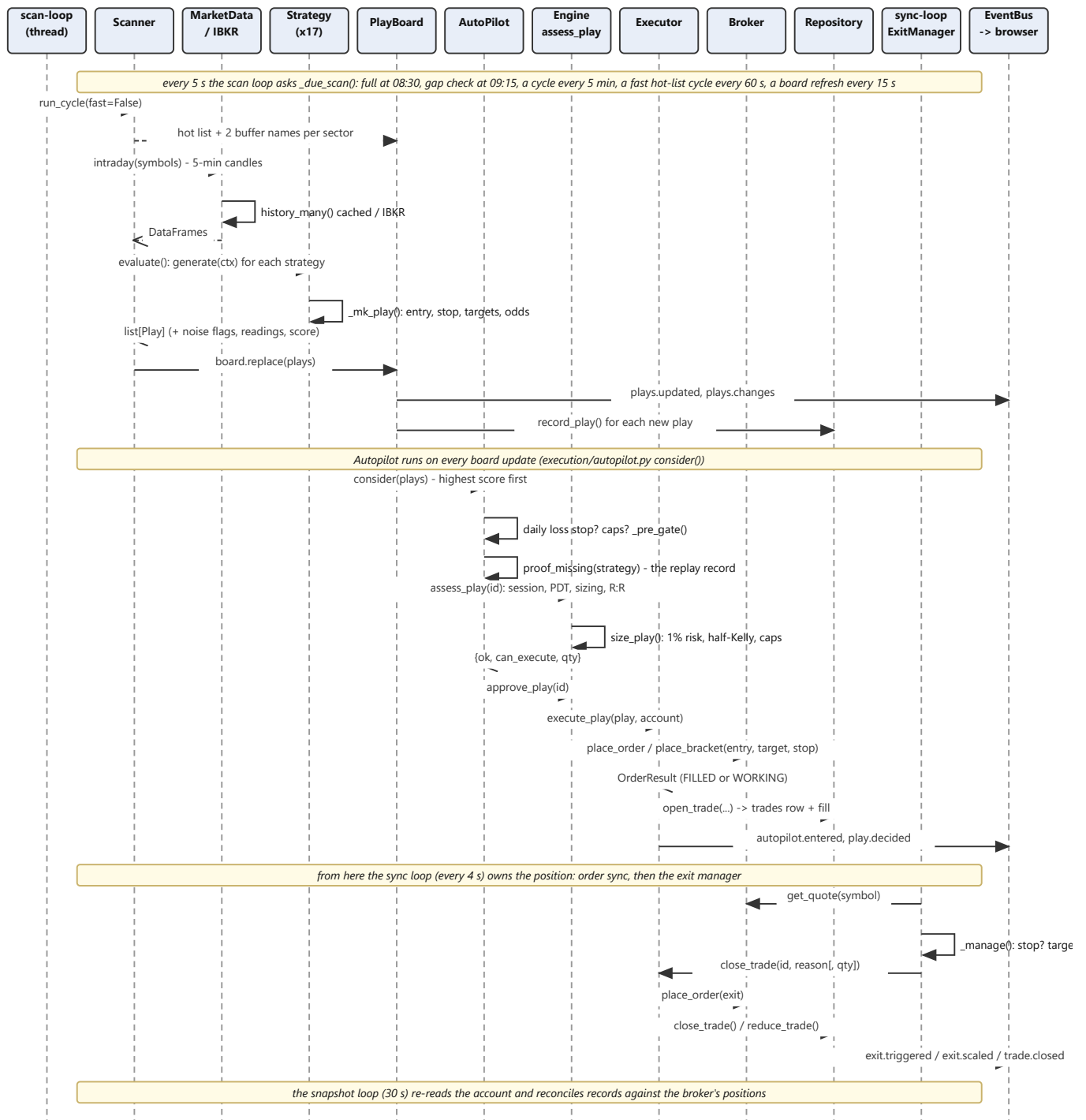


Figure 4 - Sequence: from a scan cycle to an order and, later, to the exit that closes it.

Step by step

1. **Candles.** `MarketData.intraday()` returns a pandas DataFrame per symbol (5-minute OHLCV in New York time). Daily candles come from the pickle cache in `data/bars/`, topped up from IBKR.
2. **Evaluate.** `scanner/evaluator.py evaluate()` builds a `StrategyContext` (candles, quote, activity, signals, market regime, the strategy records) and calls each enabled strategy's `generate(ctx)`. Expensive series

(ATR, VWAP, levels) are memoised on the context so seventeen strategies share one computation.

3. **Geometry guard rails.** `Strategy._mk_play()` refuses plays whose stop is too tight (below 0.6% intraday / 1.5% swing, below 0.9 ATR, or below the GARCH volatility floor) or whose reward-to-risk is outside 0.4-8. A tightened stop cannot fake a good ratio.
4. **Noise and readings.** `scanner/noise.py` flags a play that goes against the trend, against VWAP, into a gap, on a news-driven move, in a turbulent market, or on a non-trending stock (ADF/Hurst). Readings (price character, volatility forecast, next earnings) go into `evidence`.
5. **Score.** `rank_score()` = expected value in R units times the strategy weight, plus small bumps for unusual volume, a gap and enough range.
6. **Board.** `PlayBoard.replace()` keeps the top 80, notes what joined and left, and publishes `plays.updated`. Every new play is logged to `play_logs`.
7. **Autopilot.** `AutoPilot.consider()` walks the board highest score first and applies the gates in the next table. The first failing gate is the reason shown on the dashboard.
8. **Assessment and sizing.** `engine.assess_play()` checks the session, the pattern-day-trader rule (live only), and calls `risk/position_sizing.py size_play()`: risk 1% of equity per trade, lowered by the strategy's half-Kelly (a quarter of the risk until the replay has proven the strategy), the mid-day factor, the open-risk ceiling (4%), the per-position cap (12% of equity) and the per-symbol cap (15%).
9. **Order.** `Executor.execute_play()` builds a limit order (5 bps through the price) and, where the broker supports it, a native bracket with the stop and target attached. An immediate fill opens the trade; otherwise the order is tracked and `sync_open_orders()` books the fill later.
10. **Exit.** The sync loop runs `ExitManager.run_once()` every 4 s on every open trade of this venue (figure 5).
11. **A stop that outlives the app.** On IBKR the executor also keeps a good-till-cancelled stop order resting at the broker for every open trade (`execution/protective_stops.py`), made to match the trade record on every pass. It is cancelled, and the cancel confirmed, before the app sends any exit of its own; if it filled first, that fill is booked and nothing else is sent; a stop whose trade is no longer open is swept away. Beside the stop rests a limit order at the target, in one one-cancels-all group with it (IBKR reduces the other by what fills): profit is taken at the target on real prices, even on delayed quotes or with the app off, and the two can never both fill for the whole position.

The Autopilot gates, in order (execution/autopilot.py)

Gate	Rule	Setting
switched on, venue ok	Autopilot on; live trading only when the live checks pass	enabled
daily stop	no entries once today's closed trades lost 2% of equity, or gave back 30% of the day's peak gain (after +0.25%)	max_daily_loss_pct, max_giveback_pct
per-cycle / per-day / concurrent caps	1 new entry per pass, 5 per day, 10 open, 2 per strategy	max_new_per_cycle, max_auto_trades_per_day, max_auto_positions, max_per_strategy
trade type	day and swing follow the filter bar; pairs separately	trade_types
confidence	the setup's own conviction $\geq$ the floor (0.62 default; PR #22 adds a swing floor of 0.5)	min_confidence
reward-to-risk	≥ 2.0 (Aziz)	min_reward_risk
kind	valuation plays are never auto-traded	-
noise	none of the chosen flags, nor the ones the replay taught it to skip	skip_noise + learned
confirmations	a day-trade setup must appear in 2 scans in a row	min_confirmations
catalyst	optional: a news or earnings tag	require_catalyst

Gate	Rule	Setting
already in it	not holding the symbol, no entry working, not stopped out today (cooldown)	-
late in the day	no new day trade with fewer than 30 minutes to the close	min_minutes_to_close
proof	the strategy's replay record: ≥ 30 trades at $\geq +0.05R$ (also net of the stocks' own drift), a positive held-out sample of ≥ 10 trades, a reality-check p-value ≤ 0.10 across every setup tried, and costs within a third of the pre-cost edge	require_proven, proof_p_value
the learned model	in gate or size mode, and only while its own walk-forward verdict calls it usable: plays it gives under 55% are refused; shadow (the default) only logs its odds	model_mode, model_min_p
engine assessment	session valid, PDT ok, size > 0 , R:R ok, open-risk and gross-exposure ceilings	risk.*, max_gross_exposure_pct
the last look	at the live quote: refused when the spread is over $0.10R$ of the risk or the price has run $0.25R$ past the entry; within that the limit is priced off the quote; a day-trade entry unfilled after 10 minutes is cancelled	execution.max_spread_r, max_chase_r, entry_timeout_min

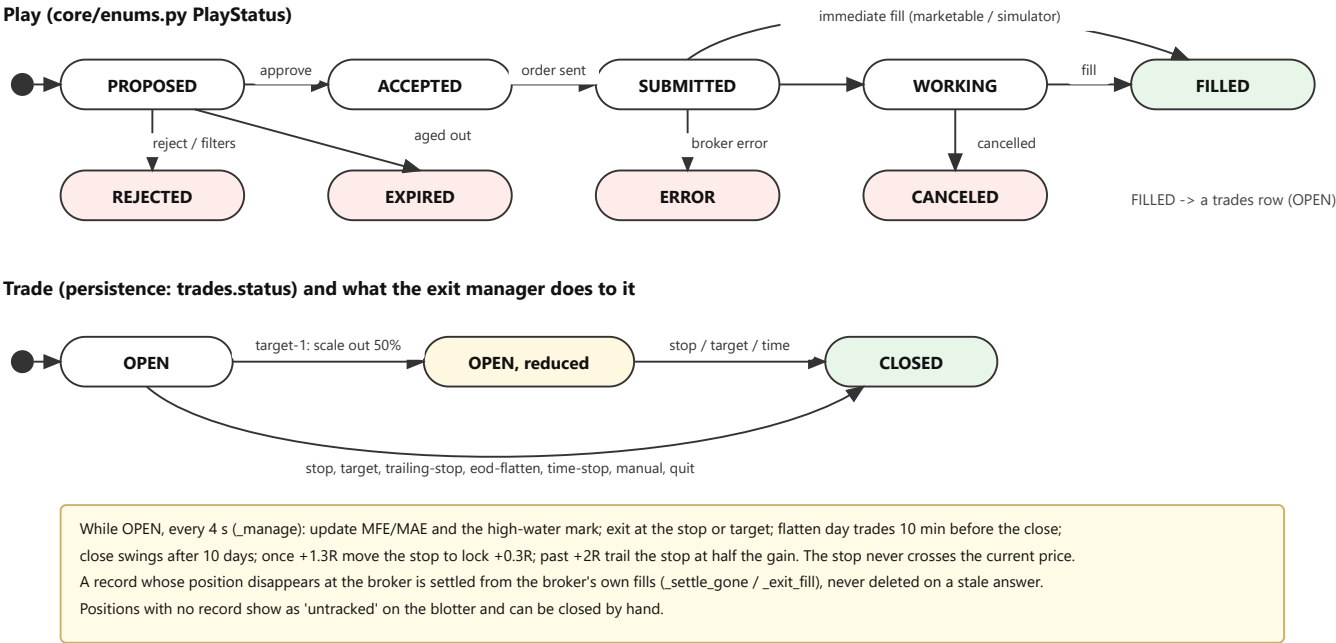


Figure 5 - State machines: a play's status and a trade's life under the exit manager.

5. The strategy catalogue

A strategy is a class with a `key`, a `kind` (technical or fundamental), a `timeframe` (INTRADAY = day trade, SWING = multi-day), a `style` (momentum trades with the move, reversal fades it, value ignores price action) and a time-of-day profile that scales confidence by the session hour. Each is switched on and weighted in the Strategies panel; the weight multiplies the rank score, never the geometry. The registry (`strategies/registry.py`) builds them from `config.yaml` plus the dashboard overrides.

Key	Title	Timeframe	Style	Idea in one line	Book
opening_range_breakout	Opening-Range Breakout	day	momentum	break of the first N minutes' range on volume	Aziz
vwap_reclaim	VWAP Reclaim / Loss	day	reversal	price crosses back over the volume-weighted average price	Aziz
ema_pullback_trend	Trend Pullback to Moving Average	day	momentum (trend)	pullback to the 9/20 EMA in a trend	Aziz
gap_and_go	Gap & Go	day	momentum	a gapper that holds its gap and pushes on	Aziz
abcd_pattern	ABCD Pattern	day	momentum	the A-B-C-D continuation	Aziz
bull_bear_flag	Bull / Bear Flag	day	momentum (open)	a pole and a tight flag in the first hour	Aziz
red_to_green	Red-to-Green (prior-day close)	day	reversal	crossing yesterday's close	Aziz
intraday_reversal	Top / Bottom Reversal	day	reversal	an extended move, a reversal candle, entry on the next candle's new high/low	Aziz
sr_bounce	Horizontal Support / Resistance	day	reversal	a bounce off a clustered level	Murphy, Aziz
rsi2_mean_reversion	RSI(2) Mean Reversion	swing	reversal	a 2-period RSI extreme in a trend	Chan
bollinger_fade	Bollinger Band Fade	swing	reversal	a close outside the band, fade back to the mean	Murphy
atr_channel_breakout	Keltner / ATR Channel Breakout	swing	momentum	a close outside the ATR channel	Murphy
week52_breakout	52-Week High/Low Momentum	swing	momentum	a new 52-week high on projected volume	Murphy
divergence_reversal	Oscillator Divergence at a Level	swing	reversal	RSI/MACD divergence at support or resistance	Murphy
failure_test	Failure Test (spring / upthrust)	swing	reversal	a probe through a swing level that closes back inside; the stop goes just beyond the test	Grimes
trend_pullback	Pullback After a Momentum Thrust	swing	momentum	the first shallow, quiet pullback to the 20-EMA after a close outside the 2.25-ATR Keltner channel	Grimes

Key	Title	Timeframe	Style	Idea in one line	Book
double_bottom	Confirmed Double Bottom / Top	swing	momentum	twin lows 2-7 weeks apart, taken only on the close beyond the peak between them; the stop sits inside the pattern	Bulkowski, Murphy
gap_reversion	Gap reversion	day	reversal	buy-on-gap: an overnight gap against the trend reverts	Chan (Algorithmic)
earnings_drift	Post-earnings drift	day	momentum	drift after an 8-K earnings surprise	Chan (Algorithmic)
relative_value_comps	Relative Value vs. Peers	swing	value	cheap vs the sector's median multiples	Pignataro
dcf_fair_value_gap	DCF Fair-Value Gap	swing	value	price far below a discounted-cash-flow value	Pignataro
valuation_football_field	Football-Field Fair-Value Band	swing	value	price outside the band of all methods	Pignataro
insider_buying	Insider buying	swing	value	clustered open-market buys by officers (Form 4)	signals

The engine reports 17 technical/statistical/fundamental strategies loaded plus the insider one; valuation plays are shown but never auto-traded.

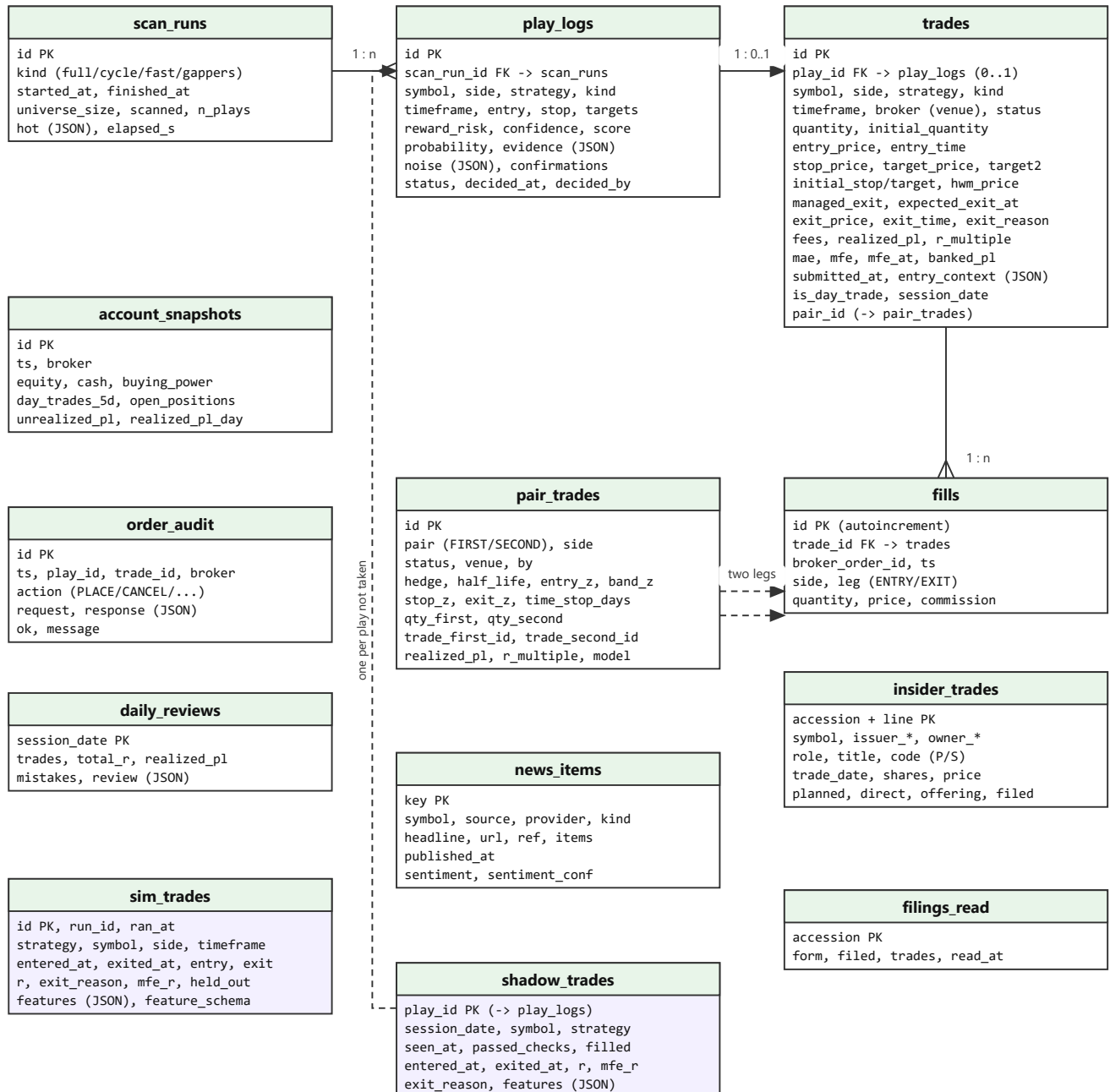
What every play carries (Play fields you will see in the API and the database)

Field	Meaning
entry, stop, targets	prices; the stop is the invalidation, targets are where profit is taken (the first target scales out half if there is a second)
reward_risk	$(\text{target} - \text{entry}) / (\text{entry} - \text{stop})$; Aziz wants ≥ 2
confidence	the setup's own conviction, 0-1, scaled by time of day
probability	the odds the play pays, calibrated against the strategy's pooled record
score	the rank on the board (expected R times weight plus bumps)
noise	list of flags such as against_trend, news_driven_move, turbulent_market, not_trending
confirmations	how many scans in a row showed this setup
evidence	a dict with the numbers behind the play: expected_r, vol_forecast, price_character, market_regime, next_earnings, a sparkline
suggested_qty, dollar_risk, notional	the sizing result
status, trade_id	the state machine of figure 6 and the trade it became

6. Data: the database, the files and the configuration

The database

`persistence/db.py` builds one SQLAlchemy engine. With no `DATABASE_URL` in `.env` it tries the MySQL URL built from the `DB_*` settings and, when that is not reachable, falls back to **SQLite at `data/autotradebot.sqlite`** - which is what runs today. `create_all()` creates the tables and `_add_missing_columns()` adds any column the ORM has and the table lacks, so a new field never needs a manual migration. Every `Repository` method opens its own short session (`session_scope()`) and returns plain dicts, so the rest of the app never touches ORM objects.



Money columns use a NUMERIC(20,6) type; JSON columns hold dicts/lists. Missing columns are added at start-up (`persistence/db.py` `_add_missing_columns()`).
Shaded: what a model learns from (research/dataset.py) - with `trades.entry_context`, the three populations of the learning guide.

Figure 6 - The database (`persistence/models_orm.py`): thirteen tables, SQLite by default.

Table	One row is	Written by
scan_runs	one scan (full, gappers, cycle, fast, plays) with its timings and counts	engine.run_scan -> repo.record_scan
play_logs	one play shown on the board, with its decision (approved, rejected, expired, auto)	repo.record_play / set_play_status
trades	one position from entry to exit, including partial exits (banked_pl), the R multiple, the play's features at the decision (entry_context) and what the fills cost: decision_price, spread_bps, entry_slippage_bps, exit_decision_price, exit_slippage_bps	repo.open_trade / reduce_trade / close_trade
fills	each broker fill, entry or exit leg, with commission	executor via the repository
account_snapshots	equity, cash, buying power every snapshot-loop pass	engine snapshot loop
order_audit	every order request and the broker's answer (JSON)	executor._audit
daily_reviews	the 16:15 review per session (JSON) plus its headline numbers	JournalOps
news_items	a headline from IBKR, SEC 8-K or Finnhub, with FinBERT sentiment	SignalStore
insider_trades / filings_read	Form 4 open-market trades and which filings were parsed	SignalStore
pair_trades	a two-leg pair position with its z-score model and both trade ids	PairDesk
sim_trades	one simulated trade of a replay run, with the play's features at the signal, a held_out flag, and what the stock's own drift made (drift_r) and costs took (cost_r) while it was held	the runner's sink -> repo.save_sim_trades
shadow_trades	one play shown and not taken, followed to its outcome on the session's candles, with its features	the 16:15 review -> repo.save_shadow_trades

Three of the columns above are the learning storage: `trades.entry_context` holds the play's features at the fill (with `submitted_at` and `mfe_at`), and the two shaded tables hold the plays not taken and the replay's trades with the same features. `research/features.py` is the one function that produces them, versioned by `FEATURE_SCHEMA`; `research/dataset.py` joins the three into one training table and `scripts/export_training_set.py` writes it as CSV. The learning guide (`docs/AutoTradeBot-Learning.pdf`) explains what is done with it.

Files under data/ (git-ignored, all rebuildable)

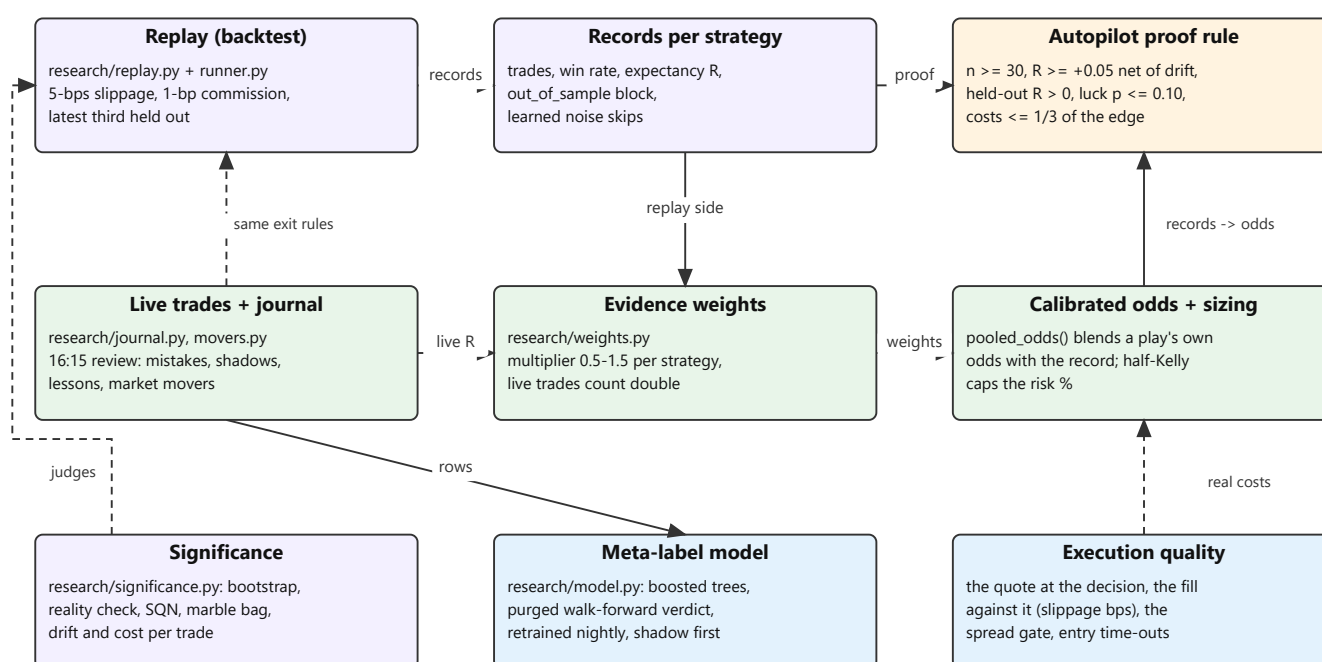
Path	Format	Holds
bars/<SYMBOL>.pkl	pickled DataFrame	one year of daily candles per stock (5,500+ files)
research/daily/<SYMBOL>.pkl	pickled DataFrame	three years of daily candles (replay.daily_years) for the stocks the replay runs on; one request per stock the first time, then topped up from bars/
research/models/	joblib + current.json	the trained meta-label models and the latest model's card
cache/	pickles + JSON	5-minute candle windows, the Nasdaq listing, SEC financials (7 days)
symbols.json	JSON	IBKR contract ids, stock types, sectors (the SymbolMaster)
watchlists/watchlist_<date>.json	JSON	the day's hot list, buffers and decisions (5 days kept)
runtime.json	JSON	every dashboard choice: mode, filters, strategies, capital, scan settings, Autopilot settings - wins over config.yaml
day_state.bin	compressed pickle	the current session so far: the board's plays, the setups settled, the gap check's pre-market levels, the last wide scan's time, the last scans' summaries - a restart picks the day up from it (engine/day_state.py)
paper_state.json	JSON	the simulator's cash, positions and round trips

Path	Format	Holds
research/replay.json, replay_runs.jsonl	JSON	the latest replay's records and every run's summary
research/training_set.csv	CSV	the exported training set (scripts/export_training_set.py)
research/intraday/	pickles	5-minute history kept for the replay
research/benchmark_spy.pkl	pickle	SPY daily returns and the fitted regime model
journal/<date>.json	JSON	the daily review with the movers report
signals/state.json, earnings_calendar.json, earnings/	JSON	signal service state and earnings dates
pairs/watch.json	JSON	the cointegrated pairs being watched

Configuration precedence

1. `.env` (secrets: IBKR host/ports/client id/account, Finnhub key, DATABASE_URL). Written only through the Connections panel by `secrets_store.py`, an allow-listed writer.
2. `config/config.yaml`: defaults for every section (`config.py` AppConfig: account, risk, scanner, strategies, valuation, execution, exit_manager, autopilot, noise, replay, journal, pairs, signals, database, app).
3. `data/runtime.json`: what you set in the dashboard. Loaded at start and applied over the YAML, so a slider you moved survives a restart and beats the file.

7. The research loop: replay, records, journal



Everything on this page is stored under `data/research/` and `data/journal/`, and shown on the Research and Reports panels.

Figure 7 - The evidence loop: the replay proves a setup, live results re-weight it, and both calibrate the odds the plays state.

Replay (`research/replay.py`). A backtest that walks candles bar by bar and lets the same strategy code propose plays as if it were live: day trades on 5-minute candles over the last 60 sessions, swings on three years of daily candles (700 sessions; each setup still sees the 300 it would see live), fills at the next bar or next open, 5 bps slippage and 1 bp commission each way, the same exit rules as the exit manager (including scale-out), news flags from the stored headlines, and the market regime as it was known that day. The latest third of sessions is **held out**: a strategy must also work on data it was not tuned on.

Which stocks, and which way in. Day-trade setups only ever see the morning's hot list, so each past session is replayed on the stocks the scan would have picked *that morning* (`research/in_play.py`: the 40 hottest by daily heat as of the session before, plus the 10 biggest opening gaps) - not on today's hot list, which is hot because of what it has just done and hands a momentum setup its own hindsight. Each day setup is followed twice: entered on sight (`entry_rule` first - the record of all trades) and entered once it has shown two bars in a row (second - how Autopilot enters when it asks for two scans running); Autopilot's record, the one the proof rule reads, uses the way in it really uses. The replay reads 5-minute candles with flat costs: it is good at throwing out a setup that loses, and a setup it likes still has to show it on paper.

Runner (`research/runner.py`). Splits the work into jobs (one stock, or ten sessions of one stock for day trades), runs them on a `ProcessPoolExecutor` with every core but two, and merges the simulated trades into per-strategy records: trade count, win rate, expectancy in R, profit factor, worst drawdown, and the same block for the held-out sample. It also learns which noise checks are worth skipping (`learned_skips`).

Proof rule. Autopilot, with `require_proven` on, refuses a strategy whose record does not show at least 30 trades averaging $+0.05R$ with a positive held-out sample of at least 10. That is why no swing trade was taken on 16 September: the records were negative or too thin. In **Live** the rule is in force whatever the setting says (`AutoPilot.proof_required`); on paper it is the user's choice, so unproven setups can be practised - at a quarter of the usual risk (`research_ops.PRACTICE_RISK`), with the settings in force kept on every trade.

Evidence weights (`research/weights.py`). Each strategy gets a multiplier between 0.5 and 1.5 from its pooled record (live trades count double), applied to the rank score. `pooled_odds()` blends a play's own stated odds with the record so the probability shown is calibrated, and `quant/sizing.py half_kelly_risk_pct()` can only lower the risk per trade.

Journal (`research/journal.py` , `movers.py`). At 16:15 the journal loop builds the day's review: the positions opened that session - closed or still open, with where the open ones stood on the close (`opened_rows()`) - every closed trade, the mistakes (entered against a flag, exited early, oversized; an entry is judged the day it is taken), the shadow outcomes of plays that were shown but not taken, per-strategy tables, and the session's biggest movers with why they moved and whether the bot traded, offered or missed them. Saved to `daily_reviews` and `data/journal/` , shown on the Reports page.

The learning storage. Everything above judges strategies as a whole. To judge one play at a time, the app keeps what every play looked like at the decision next to what happened to it, for three populations: the trades taken (`trades.entry_context`), the plays shown and not taken (`shadow_trades` , from the review) and the replay's simulated trades (`sim_trades` , from a sink on the runner). One function, `research/features.py play_features()` , describes a play with a fixed, versioned set of keys, and `research/dataset.py` joins the three. The companion guide `docs/AutoTradeBot-learning.pdf` covers the model to be trained on it, its validation and how it plugs into the gates.

Market regime (`engine/market_regime.py` , `quant/regime.py`). A two-state Markov switching model fitted by EM on three years of SPY daily returns gives the probability the market is in its turbulent state today. It feeds the `turbulent_market` noise flag and the header pill.

8. The dashboard and the API

The browser loads `index.html`, which imports `js/main.js`. Each panel is a module with an `init()`; `state.js` holds the shared state and a tiny pub/sub (`on(topic)`, `emit(topic)`); `events.js` opens the WebSocket, and every event either patches the state or triggers a fetch. Nothing is computed in the browser that the engine already knows: the engine's `snapshot()` is the single source of truth.

Module	Panel / job
topbar.js, ui.js, tooltips.js	the header pills (connection, market session, regime, equity), the drawer and tabs, hover help
plays.js, chart.js, sheet.js	the board of plays, the detail panel with the candle chart and exit routes
autopilot.js	Autopilot on/off, dry run, types, the confidence / R:R / loss-stop sliders
filters.js, strategies.js, settings.js	the filter bar (sides, timeframes, sectors), the strategy catalogue with weights in plain words, the scan schedule
blotter.js, orders.js, notes.js	open and closed trades, untracked shares, working orders, Autopilot notes
scan.js, watchlist.js	the scan status bar and the hot list / buffers / decisions
reports.js, journal.js, movers.js	the Reports page: reviews, movers, strategy tables
signals.js, pairs.js, connections.js, quit.js	the Signals page, the pairs desk, the Connections drawer (routing, .env), the quit dialog

REST routes (server/app.py)

Route	Method	Engine call
/api/state	GET	snapshot(): account, positions, autopilot, filters, market, scan, pnl, venue ...
/api/plays, /api/plays/{id}/assess approve reject chart	GET/POST	current_plays(), assess_play(), approve_play(), reject_play(), play_chart()
/api/trades, /api/trades/{id}/close managed record, /api/trades/close-all	GET/POST	trade lists, close_position(), set_trade_managed(), trade_record(), close_all_positions()
/api/positions/untracked/{symbol}/close	POST	close_untracked()
/api/orders, /api/pnl, /api/account/refresh	GET/POST	active_orders(), pnl, refresh_account_now()
/api/autopilot, /api/filters, /api/strategies[{key}]/reset	GET/POST	set_autopilot(), set_filters(), set_strategy(), reset_strategies()
/api/scan, /api/settings, /api/watchlist	GET/POST	request_scan(kind), set_scan_settings(), watchlist_state()
/api/replay[/history], /api/journal[{session}]/review, /api/signals[...]	GET/POST	start_replay(), replay_state(), journal_review(), review_session(), signals_state(), check_signals()
/api/pairs, /api/pairs/enter, /api/pairs/trades/{id}/close, /api/pairs/chart	GET/POST	pairs_state(), enter_pair(), close_pair(), pair_chart()
/api/mode, /api/paper/reset, /api/capital[/split]	POST	set_mode(), reset_paper(), set_capital(), set_capital_split()
/api/setup/secrets/reconnect/ibkr/test/paper-platform, /api/quit	GET/POST	same-machine only: setup_state(), save_secrets(), reconnect(), probe_ibkr(), begin_quit()
/ws	WebSocket	a queue on the EventBus; first message is a full snapshot

Events on the bus (core/eventbus.py)

`engine.started`, `account.snapshot`, `plays.updated`, `plays.changes`, `play.decided`, `scan.started`, `scan.failed`, `watchlist.updated`, `orders.updated`, `trade.closed`, `trades.removed`, `exit.triggered`, `exit.scaled`, `exit.stop_moved`, `exit.failed`, `exit.not_held`, `autopilot.entered`, `autopilot.would_enter`,

autopilot.skipped, autopilot.blocked, autopilot.daily_loss, autopilot.config, broker.connected /
disconnected / reconnected / down / switched, positions.mismatch, position.earnings_ahead,
journal.updated, pairs.updated, capital.updated, filters.updated, strategies.updated,
settings.updated, quit.requested / started / progress / done, engine.disarmed.

9. The functions that matter most

If you read only thirty things in the code, read these. Paths are relative to `tos_bot/`.

File	Function(s)	What it does
engine/engine.py	<code>TradingEngine.start()</code>	binds the broker, reads the account, launches the seven loops
engine/engine.py	<code>_due_scan()</code>	decides which scan is due (full / gappers / cycle / fast / plays)
engine/engine.py	<code>_run_scan(kind)</code>	runs it, records the scan_runs row, replaces the board, notes changes, hands the board to Autopilot
engine/engine.py	<code>snapshot()</code>	the dict the dashboard renders from; everything else is derived from it
engine/engine.py	<code>assess_play(id)</code>	session validity, PDT, sizing, arm state, R:R - the yes/no before any order
engine/engine.py	<code>approve_play(id)</code>	executes a play and books the trade
engine/engine.py	<code>_reconcile_open_trades()</code>	compares records with the broker's positions; settles a gone position from its fills (<code>_settle_gone</code> , <code>_exit_fill</code>); reports size mismatches
engine/engine.py	<code>_refresh_account()</code>	reads the account; keeps the last snapshot when IBKR is slow (a slow answer never reads as an empty account)
scanner/scanner.py	<code>Scanner.run_full()</code> / <code>run_cycle()</code> / <code>run_gappers()</code>	the three scans (section 3)
scanner/heat.py	<code>daily_metrics()</code> , <code>rank_by_daily_heat()</code> , <code>intraday_metrics()</code> , <code>rank_gappers()</code>	how a stock earns attention: relative volume, move in ATRs, distance to the 20-day extreme, range, dollar volume
scanner/watchlist.py	<code>DayWatchlist.apply_cycle()</code>	adopt a buffer name into the hot list, keep it waiting, or drop it
scanner/evaluator.py	<code>evaluate()</code>	context, strategies, noise flags, readings, score, signal nudges
strategies/base.py	<code>StrategyContext</code>	memoised indicators shared by all setups on one stock
strategies/base.py	<code>Strategy._mk_play()</code>	builds a Play with the geometry floors and the Douglas-style explanation
strategies/base.py	<code>calibrated_probability()</code>	blends the setup's odds with the pooled record
scanner/noise.py	<code>context_flags()</code> , <code>quant_flags()</code> , <code>event_flags()</code>	the noise checks from the books
scanner/filters.py	<code>expected_r()</code> , <code>rank_score()</code>	a play's value per unit of risk, and its rank
risk/position_sizing.py	<code>size_play()</code>	shares from the risk budget, with every cap and its label
execution/autopilot.py	<code>AutoPilot.consider()</code> , <code>_pre_gate()</code> , <code>proof_missing()</code> , <code>daily_loss_reason()</code>	the gates of section 4
execution/executor.py	<code>execute_play()</code> , <code>close_trade()</code> , <code>sync_open_orders()</code> , <code>adopt_working_orders()</code>	orders in and out, fills booked, orders already at the broker followed after a restart

File	Function(s)	What it does
execution/exit_manager.py	ExitManager.run_once() -> _manage()	stop, target, scale-out, EOD flatten, time stop, break-even, trailing (figure 5)
execution/protective_stops.py	_protect_positions(), _stand_down(), _watch_stops(), _book_target_fill()	the stop and the target resting at IBKR for each open trade, in one one-cancels-all group: placed, moved with the record, rebuilt after the scale-out, stood down before the app's own exits, booked when they fill, swept when their trade is gone
execution/order_builder.py	plan_order(), build_entry_order()	which order type and session a play may use; the limit offset
brokers/ibkr_adapter.py	IbkrBroker (get_account, history_many, place_bracket, get_fills, news_headlines)	everything IBKR, on one asyncio loop thread with timeouts
brokers/paper_adapter.py	PaperBroker.place_order(), poll()	the simulator's fills on real prices
execution/autopilot.py	kind_slots(), _kind_full(), settings_changed()	Autopilot's positions and daily entries divide between day and swing trades the way the trading capital's slider does; a settings change hands back the plays it had refused for the day
engine/engine.py	_replay_after_full_scan()	the morning full scan finishing starts the day's replay (replay.daily): once a session, remembered across a restart
persistence/repository.py	open_trade(), close_trade() - entry_latency_s / exit_latency_s	the seconds from an order going out to its fill coming back, kept per trade for the training set and the daily review
engine/engine.py	_settings_changed()	called by every live setter (split, capital, filters, strategies, Autopilot, account switch, a finished replay): re-sizes and republishes the plays, publishes Autopilot's state, pulls the board re-check forward - and never places an order itself
engine/day_state.py	DayStateOps._save_day(), _restore_day()	saves the board and the scans' state as the day goes; start() picks up the current session's, never a price
engine/reconcile.py	PositionCheck.gone(), share_counts()	when a missing position counts as gone (fresh account, settled connection, grace, two misses)
persistence/repository.py	open_trade(), reduce_trade(), close_trade(), trade_record(), pnl_summary()	the trade ledger and its R multiples
research/in_play.py	metrics_history(), in_play()	the day-trade replay's stocks: each past session's hottest by daily heat as of the session before, and its biggest gaps - the morning scan, point in time
research/history.py	IntradayHistory.load(), load_days()	5-minute candles kept on disk; load_days fetches only a stock's sessions in play and the four before each
research/replay.py	replay_intraday(), replay_swing(), summarize(), learned_skips()	the backtest and its records
research/runner.py	ReplayRunner.start(), replay_job(), session_chunks()	the parallel replay

File	Function(s)	What it does
research/weights.py	evidence_multiplier(), pooled_odds()	how records change scores and odds
research/features.py	play_features()	a play as one flat, versioned row of features - the same for live, shadow and replayed plays
research/dataset.py	training_rows(), write_csv()	the training set joined from the three populations
persistence/repository.py	save_sim_trades(), save_shadow_trades()	the learning tables' writers
research/journal.py	build_review(), opened_rows(), find_mistakes(), shadow_outcomes(), lessons()	the daily review: positions opened, trades closed, mistakes, plays not taken, lessons
pairs/desk.py	PairDesk.refresh(), watch(), enter(), manage(), close()	the pairs desk: both legs, the z-score bands, the time stop
quant/*	adf(), hurst(), half_life(), fit_garch11(), fit_markov_switching(), johansen(), best_band()	the books' models in NumPy
util/clock.py	market_status(), current_session(), time_of_day(), next_trading_day()	the NYSE calendar and sessions everything is timed by
server/app.py	create_app()	routes to engine methods; the WebSocket fan-out

10. Trading terms, explained for a computer scientist

Term	Meaning in this app
Candle / bar (OHLCV)	one time bucket of prices: open, high, low, close and volume. Daily candles for swings, 5-minute candles for day trades. Stored as pandas DataFrames indexed by New York time.
Long / short	long = buy first, profit if the price rises; short = sell borrowed shares first, profit if it falls. Side.sign is +1 / -1 and every formula multiplies by it.
Entry, stop, target	entry: the price the plan wants; stop: the price at which the idea is wrong and the position is closed; target: where profit is taken. The stop is a sunk-cost guard, not a prediction.
R (risk unit), R multiple	$R = \text{entry} - \text{stop} $ per share, the amount risked. A trade that gains 2R made twice what it risked. Expectancy in R is the average R per trade, the one number that compares strategies.
Reward-to-risk (R:R)	$(\text{target} - \text{entry}) / (\text{entry} - \text{stop})$. Aziz requires ≥ 2 ; the ranking uses expected value instead so a tightened stop cannot game it.
Position sizing, the 2% rule	$\text{shares} = (\text{equity} \times \text{risk}\%) / R$. Risk 1% per trade here (Aziz allows 2%), at most 4% open across positions.
Half-Kelly	the Kelly criterion gives the bet fraction that maximises log growth from win rate and payoff; half of it is the conventional safe version. It can only lower the configured risk %.
Slippage, commission, bps	the cost of a fill vs the quoted price, and the broker's fee. 1 bp = 0.01%. The replay charges 5 bps slippage and 1 bp commission on every fill.
Bracket order	an entry order with a stop order and a target order attached; the broker cancels one when the other fills. IBKR supports it natively.
Day trade vs swing	a day trade opens and closes the same session (and counts against the pattern-day-trader rule in a small live account); a swing is held days.
PDT rule	US rule: under \$25k equity, at most 3 day trades in 5 sessions. risk/pdt_guard.py enforces it in live mode.
VWAP	volume-weighted average price since the open: the day's fair price for institutions. Being above it is bullish context.
ATR	average true range: the typical candle range over 14 bars. Used as a volatility yardstick for stops and floors.
EMA, RSI, MACD, Bollinger, Keltner, ADX	standard indicators in indicators/ta.py: smoothed price, momentum oscillators, volatility bands, a trend-strength measure.
Support / resistance	price levels where reversals clustered before (analysis/levels.py clusters swing highs and lows).
Gap, gapper, RVOL	a gap is today's open far from yesterday's close; a gapper is a stock gapping pre-market on volume; RVOL is today's volume relative to the average - attention.
Mean reversion vs momentum	two regimes of price behaviour. ADF / Hurst / variance-ratio tests (quant/stationarity.py) say which one a series is in; reversal setups are refused on a trending series and vice versa.
Half-life	for a mean-reverting series, the time for a deviation to halve (from an AR(1) fit). Sets the expected hold of a reversal trade.
GARCH(1,1), EWMA	volatility models where today's variance depends on yesterday's shock and variance. The forecast sets a stop floor while volatility is rising.
Regime (Markov switching)	a hidden two-state model (calm / turbulent) fitted to SPY returns; the filtered probability of the turbulent state is the market pill.
Cointegration, pairs, z-score	two stocks whose price spread is stationary. The spread's z-score says how stretched it is; enter at the band, exit at the mean, stop past the stop band. Engle-Granger and Johansen are the tests.
Held-out (out of sample)	the latest third of the replay's sessions, judged separately: a setup must work on data it was not chosen on.
Expectancy, profit factor, drawdown	average R per trade; gross wins / gross losses; the worst peak-to-trough run of losses.
Paper vs live	paper = a practice account with real prices and no money (IBKR's, or the built-in simulator). Live = real money on port 4001.

Term	Meaning in this app
Scale-out	sell half at the first target and move the stop to break-even; let the rest run to the second target.
MFE / MAE	maximum favourable / adverse excursion: the best and worst the trade went while open.
Form 4, 8-K, EDGAR	SEC filings: insider trades (Form 4), material events including earnings (8-K item 2.02), and EDGAR is the SEC's free API for them.

Generated from the code as of 16 September 2026 (master plus the learning storage). When the code changes, the README and this guide should be updated together.